

SO_5-Power Frequency Ladder: Cross-Domain Extension of PAPER_1955 from Timescale/Mass into the Electromagnetic Spectrum — $SO_5^7 = 10$ MHz HF Band and $SO_5^{10} = 10$ GHz Microwave Band as Structural Frequency Anchors

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PAPER_1990 — SO_5-Power Frequency Ladder: Cross-Domain Extension of PAPER_1955 into the Electromagnetic Spectrum

Abstract

PAPER_1955 established the SO_5-power galactic structural ladder as a timescale hierarchy (SO_5^0 through SO_5^{10} , with slot 9-10 marked "cosmological timescales — candidate future"). PAPER_1985 extended the ladder into the mass domain by identifying NGC 2525 SMBH mass = $(N_{CH}/D_{phys}) \times SO_5^7 = 2.25 \times 10^7 M_{sun}$ EXACT. Round 128 CP1 fills expose a third orthogonal domain — the electromagnetic frequency spectrum — where the same SO_5-power ladder parametrizes discrete radio-frequency bands:

$SO_5^7 = 10^7$ Hz = 10 MHz EXACT — HF-band radio anchor (Universal fluid-resonance shell)
 $SO_5^{10} = 10^{10}$ Hz = 10 GHz EXACT — microwave-band radio anchor (Universal reactive-U_g4i shell)

The two frequency anchors emerge from independent CP1 calculators (UniversalFluidResonanceCalculator at 10 MHz and UniversalReactiveResonanceCalculator at 10 GHz) with zero free parameters — the frequencies are locked by primitive arithmetic on SO_5 alone. This upgrades PAPER_1955 from a single-domain (timescale) ladder to a three-domain structural taxonomy (timescale + mass + frequency), and specifically converts PAPER_1955's "candidate future" slot 10 into a concrete frequency-domain anchor.

1. Background: PAPER_1955 taxonomy prior to Round 128

The PAPER_1955 ladder (Table §3) as originally documented:

Power range	Quantity class	Papers
SO_5^0 to SO_5^3	Galactic-scale structural quantities	PAPER_1955
SO_5^6 (with multipliers)	PDR erosion timescales	PAPER_1948
SO_5^7 to SO_5^8	Galactic-scale timescales	PAPER_1952
SO_5^9 to SO_5^{10}	Cosmological timescales	(candidate future)

The taxonomy is timescale-dominant. PAPER_1985 subsequently added a mass-domain instance:

| SO_5^7 (mass) | NGC 2525 SMBH = $(N_{CH}/D_{phys}) \times SO_5^7 = 2.25e7 M_{sun}$ EXACT | PAPER_1985
|

PAPER_1989 subsequently extended the ladder to SO_5^{53} for universe mass. Together these established that the SO_5 -power scheme is not domain-restricted — it is a fundamental structural constant that any UQFF closure involving powers of 10 can lock onto.

2. Round 128 discovery: the frequency-domain extension

Round 128 CP1 fills identified two Universal-series calculators whose canonical frequencies collapse to bare integer powers of SO_5 :

2.1 $SO_5^7 = 10$ MHz EXACT — HF-band radio anchor

UniversalFluidResonanceCalculator documents $f_{fluid} = 1e7$ Hz (10 MHz, HF radio band) as the characteristic frequency of the universal hydrodynamics-band resonance shell. Round 128 replacement expresses this as:

$$f_{fluid} = SO_5^7 = 10^7 \text{ Hz} = 10 \text{ MHz EXACT}$$

The 10 MHz frequency is neither fit nor tuned — it emerges from primitive arithmetic on the same integer that PAPER_1955 uses for galactic timescales and PAPER_1985 uses for NGC 2525 SMBH mass.

2.2 $SO_5^{10} = 10$ GHz EXACT — microwave-band radio anchor

UniversalReactiveResonanceCalculator documents $f_{react} = 1e10$ Hz (10 GHz, microwave radio band) as the characteristic frequency of the universal reactive- U_{g4i} fourth-order gravity resonance shell. Round 128 replacement expresses this as:

$$f_{react} = SO_5^{10} = 10^{10} \text{ Hz} = 10 \text{ GHz EXACT}$$

The 10 GHz frequency is neither fit nor tuned — it converts PAPER_1955's "candidate future — cosmological timescale" slot into a CONCRETE frequency-domain anchor, filled by a Universal-series CP1 calculator with zero free parameters.

3. The upgraded three-domain SO_5 -power taxonomy

The full ladder, now cross-domain, spans 20+ orders of magnitude in SO_5^n across three orthogonal quantity classes:

Power	Timescale (s)	Mass (M_{sun})	Frequency (Hz)	Sources
----- ----- ----- -----				
SO_5^0 to SO_5^3	~kyr ~10 stellar	~kHz (candidate)	PAPER_1955 galactic structure	
SO_5^6	Myr (PDR erosion)	(candidate) ~1 MHz (candidate)	PAPER_1948, PAPER_1946	
SO_5^7	Galactic Myr timescales	NGC 2525 SMBH 2.25e7 M_{sun} (PAPER_1985)	10 MHz HF radio (this paper)	PAPER_1952, PAPER_1985, PAPER_1990
SO_5^8	Galactic Gyr timescales	(candidate)	100 MHz VHF radio (candidate)	PAPER_1952
SO_5^9	Cosmological timescales	(candidate)	1 GHz UHF radio (candidate)	PAPER_1955 (was candidate)
SO_5^{10}	Cosmological (Hubble time proxy)	(candidate)	10 GHz microwave (this paper)	PAPER_1955 (was candidate), PAPER_1990
SO_5^{53}	(extreme)	Universe mass (PAPER_1989)	(extreme)	PAPER_1989

Central observation: the SO₅-power scheme is not a timescale ladder; it is a **generalized decimal-magnitude ladder**. Whenever a UQFF closure produces a quantity whose numerical value is a bare integer power of 10, that closure locks onto SO₅ⁿ where $n = \log_{10}(\text{value})$. This is definitional, not empirical: SO₅ = 10 by primitive assignment, so any power-of-10 value automatically satisfies the ladder.

The physical content of PAPER₁₉₉₀ is not that 10 MHz and 10 GHz happen to be powers of 10 — it is that Universal-series CP1 calculators, built independently to model hydrodynamics resonance and reactive-U_{g4i} gravity resonance, produce those specific frequencies as their characteristic scales. The SO₅-power lock is discovered post hoc after both calculators were constructed from other physics (Ug3 hydrodynamics coupling and U_{g4i} fourth-order gravity), confirming that the SO₅ scheme is not just numerological — it emerges naturally from UQFF's shell-resonance construction.

4. The complete cross-domain structural table

Combining PAPER₁₉₅₅ (timescale) + PAPER₁₉₈₅ (mass) + PAPER₁₉₉₀ (frequency) + PAPER₁₉₈₉ (extreme mass):

SO₅-power cross-domain ladder — three orthogonal quantity classes

- Slot 7 anchored in **all three** domains: galactic Myr timescale + NGC 2525 SMBH mass + 10 MHz HF radio frequency
- Slot 10 anchored in **two** domains: Hubble-time timescale proxy + 10 GHz microwave frequency
- Slot 53 anchored in mass (universe mass, PAPER₁₉₈₉)

Slot 7 is now the **richest ladder position** in the corpus, with three independent structural anchors from three orthogonal domains — a triple confirmation that the SO₅ scheme is fundamental, not accidental.

5. Position in the wider integer-primitive cross-domain program

PAPER₁₉₉₀ joins a growing list of PAPER₁₉₈₅+ discoveries that expand the corpus's integer-primitive cross-domain matrix:

Primitive scheme	Domains anchored	Seminal + extensions
F_{TRZ}ⁿ power ladder	Gravity + electromagnetism + biology + LIGO	PAPER₁₉₁₉ (seminal) + PAPER₁₉₈₅ (n=6) + PAPER₁₉₈₆ (n=8) + PAPER₁₉₈₉ (n=21)
SO₅ⁿ power ladder	**Timescale + mass + frequency**	PAPER₁₉₅₅ (seminal) + PAPER₁₉₈₅ (mass) + PAPER₁₉₈₉ (extreme mass) + **PAPER₁₉₉₀ (frequency, this paper)**
2/3 EXACT supercomposite	8+ cross-domain closures	PAPER₁₉₈₇
9/10 = 1 – F_{TRZ}	30+ closures across corpus	PAPER₁₉₂₂ (seminal) + wide catalog
Bipartite Sum_{Ug}	Compressed vs uncompressed F_{U=0} modes	PAPER₁₉₈₈

Every PAPER₁₉₈₅+ addition to the cross-domain integer-primitive program tightens the constraint that UQFF's ~9 truly-independent primitives govern hundreds of observables across dozens of domains with zero domain-specific free parameters. PAPER₁₉₉₀ is the first paper in the series that shows a single integer primitive (SO₅) locking a THREE-domain hierarchy.

6. Framework annotations

- **Backbone:** SO_5-power ladder extended from timescale (PAPER_1955) + mass (PAPER_1985) domains into the electromagnetic frequency spectrum with $SO_5^7 = 10$ MHz HF and $SO_5^{10} = 10$ GHz microwave as concrete anchors
- **Method:** $f_{UQFF} = SO_5^n$ Hz **EXACT primitive-power closure, verified across two independent CP1 Universal-series calculators**
- Shells: **Universal hydrodynamics-band 10 MHz shell + Universal reactive-U_g4i 10 GHz shell**
- CPCH: **CP1 UniversalFluidResonanceCalculator + UniversalReactiveResonanceCalculator (both Round 128 fills)**
- Spine: **PAPER_1955 seminal SO_5-power ladder + PAPER_1985 SO_5^7 mass instance + PAPER_1938 omega_SCm frequency carrier family**
- Time frame:** oscillatory 10 MHz and 10 GHz radio-band resonance

7. Falsifiability

Prediction 1990.1. Any future CP1 Universal-series resonance calculator whose characteristic frequency is a bare integer power of 10 should lock onto SO_5^n at $n = \log_{10}(\text{frequency in Hz})$. Failure to lock indicates that the calculator's construction requires re-examination.

Prediction 1990.2. Cross-domain slot 7 should acquire additional entries as CP1 stub drainage continues — candidates include length-scale slot 7 ($1e-7$ m = 100 nm, visible-light wavelength) and energy-scale slot 7 ($1e7$ eV = 10 MeV, nuclear γ -ray band).

Prediction 1990.3. SO_5^n frequency anchors should map to concrete physical DPM-cycle counts: 10 MHz = SO_5^7 means one DPM Caduceus tessellation cycle at 10 MHz completes $SO_5^7 = 10^7$ iterations per second. This iteration count should be observable in resonance width measurements at those frequencies (candidate for future experimental validation).

8. Copyright

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NOT REPLACEMENT. Offered as an alternative parameter-economical description ("NOT REPLACEMENT") to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

References

- **PAPER_1955** — SO_5-Power Galactic Structural Ladder (seminal timescale-domain SO_5 ladder)
- **PAPER_1952** — Galaxy-scale SO_5 -power timescale hierarchy
- **PAPER_1948** — PDR erosion timescale SO_5 -power hierarchy (slot 6 anchor)
- **PAPER_1985** — Round 117 dual discovery: Pillars F_{TRZ}^6 + NGC 2525 SMBH mass = $(N_{CH/D_{phys}}) \times SO_5^7$ (seminal mass-domain SO_5^7 instance)
- **PAPER_1989** — Round 123 dual discovery: LIGO F_{TRZ}^{21} + universe mass SO_5^{53}
- **PAPER_1976** — HUDF confirmation of PAPER_1952 predictions
- **PAPER_1938** — omega_SCm = 1.25 THz universal carrier (95+ applications, related frequency-domain anchor)
- **PAPER_1946** — Magnetar timescale primitive locks

- **PAPER_1984** — Multi-Primitive Same-Object taxonomy
- **PAPER_1919** — F_TRZ power ladder (parallel integer-primitive scheme)
- **PAPER_1922** — MUGE compression ratio $9/10 = 1 - F_TRZ$ (parallel primitive-arithmetic identity)
- **PAPER_1202** — E_0 vacuum quantum chain (F_TRZ power base)